Discussion 1b Problems

Problem 1: Logic

We decide whether each equivalence is true or false and justify.

(a) $\forall x\,(P(x) \wedge Q(x)) \equiv \forall x\,P(x) \wedge \forall x\,Q(x)$

True. Suppose $\forall x\,(P(x) \wedge Q(x))$ holds. Then for any arbitrary x, both $P(x)$ and $Q(x)$ hold. Since x was arbitrary, $\forall x\,P(x)$ holds; by the same reasoning $\forall x\,Q(x)$ holds. Conversely, if $\forall x\,P(x) \wedge \forall x\,Q(x)$, then for any x both $P(x)$ and $Q(x)$ hold, giving $\forall x\,(P(x) \wedge Q(x))$. The universal quantifier distributes over conjunction.

(b) $\forall x\,(P(x) \vee Q(x)) \equiv \forall x\,P(x) \vee \forall x\,Q(x)$

False. Let $P(x)$ be "x is even" and $Q(x)$ be "x is odd" over $\mathbb{Z}$. Then $\forall x\,(P(x) \vee Q(x))$ is true since every integer is either even or odd. However, $\forall x\,P(x)$ (every integer is even) and $\forall x\,Q(x)$ (every integer is odd) are both false, so the right side is false.

(c) $\exists x\,(P(x) \vee Q(x)) \equiv \exists x\,P(x) \vee \exists x\,Q(x)$

True. Suppose $\exists x\,(P(x) \vee Q(x))$. Then there exists a specific witness x_0 such that $P(x_0) \vee Q(x_0)$ holds. Either $P(x_0)$ holds, giving $\exists x\,P(x)$, or $Q(x_0)$ holds, giving $\exists x\,Q(x)$. In either case $\exists x\,P(x) \vee \exists x\,Q(x)$ holds. The existential quantifier distributes over disjunction.

(d) $\exists x\,(P(x) \wedge Q(x)) \equiv \exists x\,P(x) \wedge \exists x\,Q(x)$

False. Let $P(x)$ be "x is even" and $Q(x)$ be "x is odd" over $\mathbb{Z}$. Then $\exists x\,P(x)$ is true (e.g. $x = 2$) and $\exists x\,Q(x)$ is true (e.g. $x = 3$), so the right side is true. However, $\exists x\,(P(x) \wedge Q(x))$ requires a single integer that is simultaneously even and odd, which is impossible, so the left side is false.

Problem 2: Contraposition

Claim: If $a + b < c + d$, then $a < c$ or $b < d$.

Proof. We prove the contrapositive: if $\neg(a < c \vee b < d)$, then $\neg(a + b < c + d)$.

By De Morgan's law, $\neg(a < c \vee b < d)$ is equivalent to $a \geq c$ and $b \geq d$. Adding these two inequalities gives:

$$a + b \geq c + d$$

which is exactly $\neg(a + b < c + d)$. Since we have proved the contrapositive, the original statement holds. $\square$

Problem 3: Perfect Square

Claim: Every odd perfect square is of the form $8k + 1$ for some integer k.

Proof. Let n be an odd perfect square. Then $n = m^2$ for some integer m. Since n is odd and $n = m^2$, m must also be odd (if m were even, m^2 would be even). Therefore we may write $m = 2j + 1$ for some integer j. Expanding:

$$n = (2j + 1)^2 = 4j^2 + 4j + 1 = 4j(j + 1) + 1$$

Since j and $j + 1$ are consecutive integers, one of them must be even, so $j(j + 1)$ is always even. Thus $j(j + 1) = 2k$ for some integer k, giving:

$$n = 4 \cdot 2k + 1 = 8k + 1$$

Therefore every odd perfect square is of the form $8k + 1$. $\square$

Problem 4: Numbers of Friends

Claim: If there are $n \geq 2$ people at a party, at least 2 of them have the same number of friends at the party.

Proof. Each person's friend-count lies in $\{0, 1, \ldots, n - 1\}$, which contains n possible values. However, the values 0 and $n - 1$ cannot both occur simultaneously: if some person knows everyone ($n - 1$ friends), then no person can have 0 friends, since they are at least friends with that person. Therefore at most $n - 1$ distinct friend-count values are achievable at any one time.

By the Pigeonhole Principle, with n people (items) and at most $n - 1$ possible friend-counts (containers), at least two people must share the same number of friends. $\square$

Problem 5: Fermat's Contradiction

Claim: $2^{1/n}$ is irrational for any integer $n \geq 3$.

Proof. Suppose for contradiction that $2^{1/n} \in \mathbb{Q}$ for some $n \geq 3$. By definition of rational numbers, we may write:

$$2^{1/n} = \frac{a}{b}$$

for some positive integers a, b. Raising both sides to the n-th power:

$$2 = \frac{a^n}{b^n} \implies a^n = 2b^n = b^n + b^n$$

This gives positive integers a, b, b satisfying $b^n + b^n = a^n$ with $n \geq 3$. By Fermat's Last Theorem, there are no positive integers a, b, c such that $a^n + b^n = c^n$ for $n \geq 3$. This is a contradiction. Therefore $2^{1/n}$ is irrational for all $n \geq 3$. $\square$